

Week 5

Deep Learning

Question: 1

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, You apply the transformation Ax and obtain a new vector.

Which of the following best describes the geometric effect of this matrix transformation on the vector x ?

Options:

- A) The vector x is scaled but remains in the same direction — no rotation occurs.
- B) The vector x is transformed into a vector orthogonal to itself.
- C) The vector x is mapped to the origin — it lies in the null space of A .
- D) The vector x is rotated to a new direction and scaled — the direction changes.

Correct Answer: D

Explanation:

$Ax = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$ is not a scalar multiple of x , so its direction has changed.

The transformation involves both rotation and scaling.

Question: 2

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, a student claims that x is an eigen vector of matrix A .

Is the student's claim correct? If so, what is the corresponding eigenvalue?

Options:

- A) Yes, x is an eigenvector of A with eigenvalue $\lambda = 2$.
- B) No, the transformation changes the direction of x , so it cannot be an eigenvector.
- C) Yes, x is an eigenvector with eigenvalue $\lambda = 3$.
- D) No, x maps to a zero vector under A , indicating it lies in the null space.

Correct Answer: C

Explanation:

$$Ax = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Since the output is a scalar multiple of x , it confirms that x is an eigenvector, and the eigenvalue is $\lambda = 3$.

Question: 3

A company models user transition between two online platforms, AppX and AppY, using a matrix:

$$M = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}$$

Each day, users switch between these apps according to the matrix. If on *day* 0, 300 users are on AppX and 200 on AppY, which of the following best describes what will eventually happen to the distribution of users?

Options:

- A) The number of users will keep oscillating without settling
- B) The users will all eventually shift to AppX
- C) The distribution will stabilize in the ratio of the dominant eigenvector
- D) The user numbers will keep increasing exponentially

Correct Answer: C

Explanation: The matrix is stochastic; over time, repeated multiplication leads to convergence to the dominant eigenvector, resulting in a steady-state distribution

Question: 4

Given matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, find the dominant eigen value of A.

Fill in the answer : _____

Correct Answer : 5

Explanation: The eigenvalues of A are found by solving $\det(A - \lambda I) = 0$, which gives eigenvalues $\lambda = 5, 2$.

Question: 5

Which of the following statements are true

Options:

- A) The eigenvectors corresponding to different eigenvalues are linearly independent.
- B) The eigenvectors of a square symmetric matrix are orthogonal.
- C) The eigenvectors of a square symmetric matrix can thus form a convenient basis.
- D) A statement which is wrong.

Correct Answer: A, B, C

Explanation:

Question: 6

Consider a sequence of vectors $v_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. If we compute $v_n = A^n v_0$, what will v_n converge to?

Options:

- A) A random vector
- B) A zero vector
- C) A multiple of the dominant eigenvector of A
- D) The vector with the smallest eigenvalue

Correct Answer: C

Explanation: The repeated application of A leads to convergence toward the dominant eigenvector's direction if the dominant eigenvalue exists and is unique.

Data for the questions from 7 to 9

A data scientist is simulating a simplified web graph with 3 webpages (A, B, C). The transition probability matrix for a web surfer moving from one page to another, where each column represents a probability distribution is modeled as:

$$M = \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0.3 & 0.5 & 0.3 \\ 0.2 & 0.3 & 0.4 \end{bmatrix}$$

The scientist initializes a random visitor distribution vector $v_0 = \begin{bmatrix} 0.3 \\ 0.4 \\ 0.3 \end{bmatrix}$.

She now repeatedly applies the transformation $v_{k+1} = M v_k$.

Question: 7

What kind of matrix is M , why is this classification important for understanding the long-term behavior of v_k as $k \rightarrow \infty$?

Options:

- A) M is an orthogonal matrix, so the vectors v_k remain unchanged in length.
- B) M is a symmetric matrix, which ensures all eigenvalues are real.
- C) M is a stochastic matrix, so it preserves probability distributions and has a dominant eigenvalue equal to 1.
- D) M is a diagonal matrix, so it converges faster under repeated multiplication.

Correct Answer: C

Explanation: A stochastic matrix has non-negative entries and columns summing to 1. Its dominant eigenvalue is 1, and repeated application will converge the vector to the dominant eigenvector.

Question: 8

Suppose after multiple iterations the vector v_k converges to: $v^* = \begin{bmatrix} 0.34 \\ 0.39 \\ 0.27 \end{bmatrix}$. Which of the following best interprets this result in the context of web ranking?

Options:

- A) Page B has the highest rank, followed by A and C, based on steady-state visit probability.
- B) Page A is most likely to be visited, so it should be assigned lowest priority in ranking.
- C) All pages are equally ranked because the initial vector had equal probability.
- D) This indicates the convergence failed, as stochastic matrices must lead to uniform distributions.

Correct Answer: A

Explanation: In steady-state, the value in each position of the vector indicates the long-term visit probability. Thus, page B is ranked highest, followed by A and then C. The initial distribution does not influence the final rank due to convergence toward the dominant eigenvector.

Question: 9

Suppose instead of starting with $v_0 = \begin{bmatrix} 0.3 \\ 0.4 \\ 0.3 \end{bmatrix}$, the scientist starts with $v_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$. Will the final ranking vector after many iterations be different?

Options:

- A) No, because the convergence behavior is governed only by the dominant eigenvalue and eigenvector.
- B) Yes, because the initial vector strongly favors page A.
- C) No, because the matrix is symmetric, so initial vector doesn't matter.
- D) Yes, because the matrix has complex eigenvalues and depends on initial phase.

Correct Answer: A

Explanation: for a matrix with a dominant eigenvalue, the sequence v_k converges to a multiple of the dominant eigenvector, regardless of the starting vector. Hence, ranking is stable.

Question: 10

A weather model predicts transitions between two states — Sunny and Rainy — using

$$A = \begin{bmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{bmatrix}, x_0 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

Here, x_0 is the initial probability distribution across states. The meteorologist applies the transformation iteratively: $x_{k+1} = Ax_k$

which of the following best describes the long-term behavior of the sequence x_k ?

Options:

- A) The sequence will vanish toward zero.
- B) The sequence will oscillate between states.
- C) The sequence will converge to a steady-state probability vector.
- D) The sequence will explode due to the growing norm.

Correct Answer: C

Explanation: Since A is a stochastic matrix, its dominant eigenvalue is 1, and it has a steady-state eigenvector corresponding to that eigenvalue.

Question: 11

A student studies the behavior of the transformation:

$$A = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.3 \end{bmatrix}, x_0 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

She computes the sequence $x_1 = Ax_0, x_2 = Ax_1 \dots$

What is the nature of this sequence as $k \rightarrow \infty$, and why?

Options:

- A) The sequence will converge to a zero vector because both eigenvalues have magnitude < 1 .
- B) The sequence will oscillate due to eigenvalue signs.
- C) The sequence will explode since the matrix has entries > 1 .
- D) The vector will converge to a steady state with unit magnitude.

Correct Answer: A

Explanation:

The eigenvalues of A are $\lambda_1 = 0.5, \lambda_2 = 0.3$, both less than 1 in magnitude. Hence, repeated multiplication causes the sequence $A^k x_0 \rightarrow 0$ as $k \rightarrow \infty$. Each coordinate decays exponentially.

Question: 12

Given three vectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, v_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$. To determine whether the set $\{v_1, v_2, v_3\}$ can serve as basis for R^3 , Which of the following steps is the most appropriate?

Options:

- A) Check if the determinant of the matrix $A = [v_1 v_2 v_3]$ is non-zero.
- B) Check if the sum $v_1 + v_2 + v_3 = 0$
- C) Verify if any vector can be written as a linear combination of the others

D) Both A and C

Correct Answer: D

Explanation: A set of vectors forms a basis iff they are linearly independent and span the space.

Question: 13

Given: $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$. What does the determinant of A tell us about the vectors?

Options:

- A) The vectors form a basis for R^3 .
- B) The vectors are linearly dependent and hence do not form a basis
- C) The vectors span R^3 but are not independent
- D) Cannot conclude anything from the determinant

Correct Answer: B

Explanation:

$\text{Det}(A) = 0 \rightarrow$ columns are linearly dependent. Hence, the set cannot be a basis.

Question: 14

Which of the following statements is always true for a square matrix $A \in R^{n \times n}$ that has distinct eigenvalues?

Options:

- A) The eigenvectors of A form a linearly independent set
- B) All eigenvectors of A are orthogonal
- C) The eigenvectors of A are linearly dependent
- D) A must be a symmetric matrix

Correct Answer: A

Explanation:

For any square matrix with distinct eigenvalues, its eigenvectors are guaranteed to be linearly independent, even if the matrix is not symmetric. They are not necessarily orthogonal unless the matrix is symmetric.

Question: 15

Why are the eigenvectors of a square symmetric matrix considered special?

Options:

- A) They are always zero vectors
- B) They are complex even when the matrix has real entries
- C) They are orthogonal to each other and can form a basis
- D) They cannot be used to diagonalize the matrix

Correct Answer: C

Explanation:

Eigenvectors of a real symmetric matrix are not just linearly independent—they are orthogonal, and thus form an orthogonal (even orthonormal) basis for R^n .

Question: 16

Let A be a real symmetric matrix. Which of the following statements is true?

Options:

- A) The vector that maximizes $x^T Ax$ under the constraint $\|x\| = 1$ is the eigen vector of A corresponding to its smallest eigenvalue.
- B) The vector that minimizes $x^T Ax$ under the constraint $\|x\| = 1$ is the eigen vector of A corresponding to its largest eigenvalue.
- C) The maximum and minimum of $x^T Ax$ under $\|x\| = 1$ are both achieved by arbitrary unit vectors.
- D) The vector that maximizes $x^T Ax$ under the constraint $\|x\| = 1$ is the eigenvector corresponding to the largest eigenvalue of A .

Correct Answer: D

Explanation:

For a real symmetric matrix A , maximizing $x^T Ax$ over unit vectors x gives the largest eigenvalue, and the maximizer is the corresponding eigenvector.

Question: 17

Which of the following statements are true regarding eigenvectors and symmetric matrices?

Options:

- A) Eigenvectors corresponding to different eigenvalues are linearly independent.
- B) Eigenvectors of any matrix are always orthogonal.
- C) Eigenvectors of a square symmetric matrix are orthogonal.
- D) Eigenvectors of a square symmetric matrix can form a basis.

Correct Answers: A, C, and D

Explanation:

eigenvectors with different eigenvalues are linearly independent.

The eigenvectors of a square symmetric matrix are orthogonal.

Since the eigenvectors are orthogonal and span the space, they form a convenient basis.

Data for the questions 18 and 19

WebTrack Inc. is analyzing session data from users of their *e – learning* platform. For each session, they log:

- x: Time spent in hours
- y: Number of video modules visited
- z: Number of clicks made during the session

The analytics team is trying to identify if the number of clicks gives any additional information beyond modules visited. Here's the revised data from a random sample of user sessions:

Session	x	y	z
1	0.95	1.2	1.2
2	0.42	0.1	0.1
3	0.30	1.3	1.3
4	0.38	1.7	1.7
5	0.48	1.1	1.1
6	0.59	2.2	2.3
7	0.65	1.4	1.3
8	0.76	0.85	0.86
9	0.71	0.95	0.96

They calculate the Pearson correlation coefficient ρ_{yz} to assess the similarity between y and z.

Question: 18

What does the correlation coefficient ρ_{yz} tell us about the relationship between modules visited (y) and clicks (z)?

Options:

- A) They are unrelated
- B) They are moderately correlated
- C) They are strongly positively correlated
- D) They are inversely related

Correct Answer: C

Explanation:

The values of y and z are almost identical in most sessions, indicating a high positive correlation (ρ close to 1).

Question: 19

How should the analytics team handle column z when building a predictive model?

Options:

- A) Drop one of y or z to reduce redundancy
- B) Keep both y and z to increase accuracy

- C) Use only z because it is slightly higher in some rows
- D) Drop both y and z since they are similar

Correct Answer: A

Explanation:

Due to high correlation, one column is redundant. Keeping both could cause multicollinearity in models.

Question: 20

You are working on a dataset with 100 features. After applying PCA, you reduce it to 2 dimensions. These new dimensions:

Options:

- A) Are linear combinations of original features and orthogonal
- B) Represent the mean of the original features
- C) Are highly correlated with each other
- D) Have lower variance than all other dimensions

Correct Answer: A

Explanation: PCA constructs orthogonal components that capture the directions of maximum variance in the data.

Data for the questions 21 and 22

A data scientist is analyzing EEG signals recorded from 100 channels. The signals are known to contain some background noise. She decides to apply Principal Component Analysis and computes the eigenvalues and eigenvectors of the covariance matrix of the data. She retains only the **top 10 eigenvectors** and discards the remaining 90 components.

Question: 21

What is the most principled justification for discarding the 90 components?

Options:

- A) The discarded components correspond to the directions where the data varies the most, so they are likely to be noisy.
- B) The retained components are likely to contain only noise, so discarding the rest ensures the signal is preserved.
- C) The discarded components represent directions with very low variance, which often contain noise rather than meaningful structure.
- D) All PCA components contribute equally to reconstruction, so discarding any of them does not affect signal quality.

Correct Answer: C

Explanation: PCA orders components by variance. Low-variance directions (i.e., small eigenvalues) are often due to noise. Discarding them helps denoise the signal with minimal reconstruction loss.

Question: 22

Which mathematical property of the eigenvectors ensures that the new PCA basis does not mix signal and noise directions?

Options:

- A) Eigenvectors of a matrix are always parallel to each other, preserving component alignment.
- B) Eigenvectors of a symmetric matrix are orthogonal, allowing projection without interference.
- C) Eigenvectors are complex-valued, making them useful for separating signal from noise.
- D) Eigenvectors are unit vectors, ensuring normalization of signal power.

Correct Answer: B

Explanation: The covariance matrix is symmetric, and its eigenvectors form an orthogonal basis, which ensures that signal and noise directions are linearly separable and do not interfere when projecting or reconstructing data.

Information for the questions 23 and 24:

A researcher is working on compressing a dataset of grayscale face images, each of size 64×64 pixels (4096 features per image). Using PCA, she finds that the top 150 eigenvectors retain 95% of the total variance.

Out of curiosity, she reconstructs some images using only the remaining 3946 components (those corresponding to the smallest eigenvalues).

Question: 23

What is the most likely visual outcome of reconstructing images using only these 3946 low-variance components?

Options:

- A) The images will appear distorted or noisy, lacking meaningful structure.
- B) The images will retain essential facial features with only minor loss in brightness.
- C) The images will be highly detailed, as fine-grained features are captured in low-variance components.
- D) The images will look the same, since orthogonality ensures perfect reconstruction from any subset of components.

Correct Answer: A

Explanation: The low-variance components contribute little to the signal's structure and often capture noise or unimportant fluctuations. Reconstructing from them leads to loss of key features and yields poor image quality.

Question: 24

Why is it mathematically sound to use only the top-k eigenvectors for dimensionality reduction in PCA?

Options:

- A) Because the top-k eigenvectors span the entire null space of the original matrix.
- B) Because eigenvectors with smaller eigenvalues are linearly dependent and redundant.
- C) Because PCA only works with orthogonal matrices of full rank.
- D) Because these directions correspond to maximum variance, preserving most information with minimal components.

Correct Answer: B

Explanation: PCA selects directions (eigenvectors) that capture the most variance (eigenvalues). Retaining the top-k ensures that most of the data variability is preserved while reducing dimensionality.

Data for the questions 25 and 26

A fintech startup is building a real-time fraud detection system based on thousands of transaction features (e.g., time, location, merchant type, device, customer history, etc.). The high dimensionality of the data is slowing down model training and inference. To resolve this, the data science team applies PCA and retains only the top 25 principal components from the original 300 features.

After transformation, they observe the following:

- The components are uncorrelated.
- The system runs much faster.
- The fraud detection performance remains largely unaffected.

Question: 25

Which of the following best explains why PCA was effective in this case?

Options:

- A) PCA minimized the variance within each individual feature, making the data easier to compress.
- B) PCA transformed the features into a new space where components are uncorrelated and high-variance directions are preserved, reducing dimensionality with minimal loss of information.
- C) PCA converted all non-linear features into linear ones, which improved model interpretability.

- D) PCA forced all features to have zero mean and unit variance, which improved classification accuracy.

Correct Answer: B

Explanation: PCA projects the data into an orthogonal space with uncorrelated components that maximize variance and retain the most informative directions, enabling reduction from 300 to 25 features while preserving most of the structure.

Question: 26

What mathematical guarantee offered by PCA ensures that the model is not negatively affected by redundant or overlapping features?

Options:

- A) PCA maximizes correlation between transformed features to preserve group structure.
- B) PCA aligns features with the original feature axes, making reconstruction error zero.
- C) PCA ensures that the covariance between the new dimensions is minimized, thereby removing redundancy.
- D) PCA clusters features into distinct groups based on their variance contribution.

Correct Answer: C

Explanation: One of the interpretations of PCA is that it produces uncorrelated components, i.e., the covariance between the new axes is zero. This orthogonalization helps eliminate redundancy from overlapping or correlated features.

Information for questions 27 and 28:

You are given a dataset of grayscale human face images. Each image is of size 100×100 pixels and the total number of images are 500. PCA is applied to reduce the dimensionality by computing the top 100 eigenvectors of $X^T X$. Each eigen vector is reshaped into a 100×100 image and is referred to as an eigenface. Each original image is now approximated using just 100 scalar coefficients corresponding to these eigenfaces.

Question: 27

What is the role of eigenfaces in compressing the face image dataset?

Options:

- A) They duplicate all faces to save space
- B) They reduce resolution of images for faster display
- C) They form a lower-dimensional basis to represent face images efficiently
- D) They are used to colorize grayscale images

Correct Answer: C

Explanation:

Eigenfaces are the principal components obtained from PCA. They help express any 10,000-dimensional face image using just a small number of weights (projections), reducing both storage and computation.

Question: 28

You have 500 images, each originally 10,000 dimensions. After projecting onto the top 100 eigenfaces, how many scalar values are needed to store all 500 compressed images?

Options:

- A) 10000
- B) 50000
- C) 5000000
- D) 600

Correct Answer: B

Explanation: Each of the 500 images is now represented using 100 numbers. So, total storage = $500 \times 100 = 50,000$ scalar values.

Question: 29

What is the correct form of the Singular Value Decomposition of a real matrix $A \in R^{m \times n}$?

Options:

- A) $A = U\Sigma V^T$
- B) $A = UDU^T$
- C) $A = V\Sigma V^T$
- D) $A = PVP^T$

Correct Answer: A

Explanation: SVD expresses A as a product of two orthogonal matrices U and V , and a diagonal matrix Σ of singular values.

Question: 30

In the SVD of a real matrix $A = U\Sigma V^T$, what is the nature of the matrix Σ ?

Options:

- A) A diagonal matrix with real values
- B) A diagonal matrix with non-negative real numbers
- C) An upper triangular matrix
- D) A symmetric matrix

Correct Answer: B

Explanation:

The singular values of A are always non-negative and are placed on the diagonal of Σ .